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CO electrolysers with 51% energy efficiency towards C_{2+} using porous separators

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Electrochemical CO_2 reduction can generate multi-carbon (C_{2+}) products via a CO_2-to-CO cascade followed by CO reduction (COR). However, COR energy efficiency remains below 40% due to sluggish ion transport within charge-selective membranes. Here we introduce an uncharged porous separator that enables facile transport of both ion types, reducing ohmic resistance and superconcentrating cations at the catalyst surface—lowering COR voltage by 150 mV at 200 mA cm^{-2}. In previous electrolyser designs, porous separators were limited by cathode-to-anode H_2 crossover; the low diffusivity of C_2H_4 and CO in water allows a separator three times thinner and 1.6 times more porous, markedly reducing overpotential. Operating at elevated temperatures with a nickel–iron-based anode further lowers voltage by 330 mV, leading to a full-cell voltage of 1.95 V at 200 mA cm^{-2} and an energy efficiency of 51% to C_{2+} products sustained over 250 h. The system also achieves a CO single-pass conversion of 97% and a C_2H_4 concentration of 87 wt% in the product gas stream.

Electrochemical CO_2 reduction (CO_2R), when powered by renewable energy, offers a means to transform waste CO_2 into valuable multi-carbon (C_{2+}) chemicals and fuels[1,2]. Although the field has advanced rapidly, high energy input presents a barrier to wider application[3,4]. CO_2R in alkaline media enables the generation of C_{2+} products[5–8]; however, under alkaline conditions, the reactant CO_2 converts to carbonate, and regenerating this to CO_2 is costly[9,10]. Performing CO_2R in acidic electrolytes, and employing bipolar membranes to regenerate CO_2, addresses the CO_2 loss problem, but the energy efficiency (EE) remains below 20% (Fig. 1a)[9,11,12].

Achieving both carbon and energy efficiency is possible through cascade CO_2R, starting with CO_2-to-CO conversion followed by CO reduction (COR). The first CO_2-to-CO step can be performed in a high-temperature solid-oxide electrolyser with an EE that exceeds 80%[13,14]. Thus, realizing the full promise of the cascade approach will require efficiency gains in the COR step.

Operating COR in alkaline conditions benefits from improved kinetics and minimal Nernstian pH losses[8,15,16]. COR is typically performed in a membrane electrode assembly (MEA) in which the cathode and anode are separated by an anion exchange membrane (AEM, Fig. 1b). The AEM provides a physical separation and controls ion, liquid and gas transport between electrodes. This has led to EEs that have reached a peak value of 40% so far towards C_{2+} products (Fig. 1a)[14,17–20]. Even with such progress in the field, the cost of (desirably low-carbon-intensity) electrolyser electricity is forecast to dominate cost (Supplementary Figs. 1 and 2 and Supplementary Notes 2 and 3).

Hydroxide transport in AEMs occurs via a hopping mechanism, where hydroxide ions move from one fixed positively charged site to

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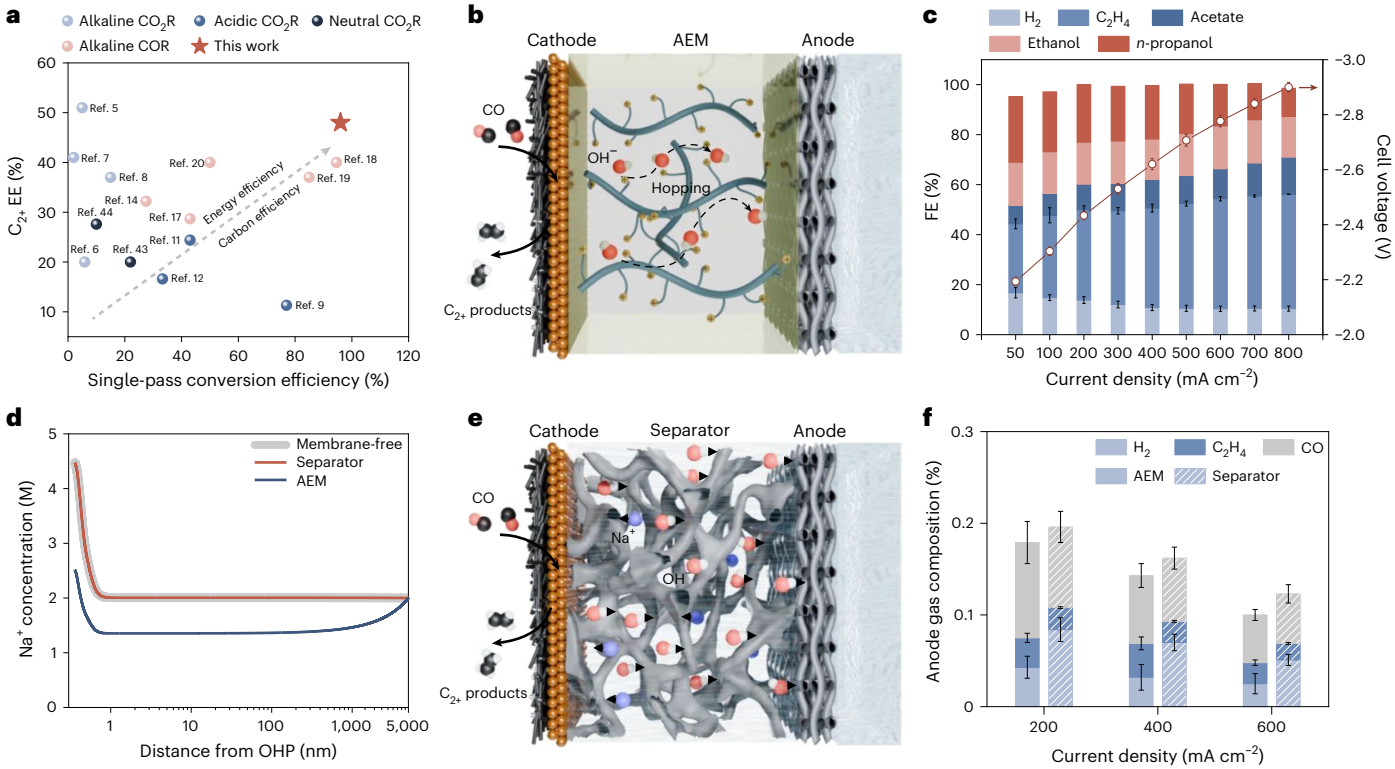

Fig. 1 | Performance of the MEA with an AEM. a, Summary of alkaline CO_2R[5–8], acidic CO_2R[9,11,12], neutral CO_2R[43,44] and alkaline COR[14,17–20] performance towards C_{2+} products. **b**, Schematic of the MEA with an AEM. **c**, Product distribution and cell voltage of the MEA with an AEM at different current densities. **d**, Cation concentration as a function of distance from the outer Helmholtz plane (OHP) of the cathode for the MEA with an AEM, a separator and the membrane-free electrolyser. **e**, Schematic of the MEA with a separator. **f**, Anode gas compositions at different current densities of the MEA with an AEM and a separator. The rest of the anode gas is O_2. Where error bars are shown, values are means, and error bars indicate s.d. ($n = 3$ replicates).

another, facilitated by periodic hydration and dehydration (Fig. 1b). This process has a kinetic limit owing to the energy required to break the electrostatic interactions between the hydroxide ions at each positively charged site, as well as the energy required for continuous hydration/dehydration[21,22]. As a result, present-day AEMs exhibit low conductivity, a challenge shared with AEM fuel cells and electrolysers[23,24].

In this work, we introduce a porous hydrophilic separator as an alternative to charge-selective membranes in COR electrolysers. This separator enables efficient ion transport while blocking gas crossover and creates a local cation superconcentration at the cathode, reducing cell voltage and improving EE. Compared with AEM-based systems, the porous-separator approach lowers full-cell voltage by 150 mV at 200 mA cm^{-2}. We further integrate a nickel–iron anode and operate at elevated temperature, achieving a full-cell voltage of 1.95 V and an EE of 51% toward C_{2+} products that was sustained over 250 h.

Limitations of AEM and membrane-free systems

To baseline AEM system performance, we performed COR using a Cu cathode (scanning electron microscopy image in Supplementary Fig. 3) and an iridium oxide (IrOx) anode separated by an AEM. When we used a highly alkaline anolyte, the C_2H_4 and C_{2+} Faradaic efficiency (FE) increased with applied current density, reaching 35% and 86% at current densities over 200 mA cm^{-2} (Fig. 1c and Supplementary Fig. 4). We measured a full-cell voltage of 2.43 V at 200 mA cm^{-2}, corresponding to an EE of 35% for C_{2+} products and 16% for C_2H_4 (Supplementary Fig. 5). This result is typical of the field[14,17–20].

In bulk electrolyte, in contrast to the AEM, ions move freely: indeed, alkaline electrolyte has three times the conductivity of the AEM[25]. To lever this advantage, we initially tested a membrane-free set-up wherein the electrodes were separated solely by a layer of bulk electrolyte (Supplementary Fig. 6). However, we observed substantial cathode gas crossover to the anode compartment, losing over 20% of the C_2H_4 to the anode (Supplementary Figs. 7 and 8). Additionally, we detected O_2 generated by the anodic oxygen evolution reaction (OER, corresponding to ~10% of the O_2 produced) in the cathodic gas outlet. It is expected that gas bubbles generated at the electrode surface rise through the central electrolyte via buoyancy-driven convection and, in the absence of a membrane, mix with the cathode products. Increasing the interelectrode thickness reduced gas crossover but substantially raised cell voltage (Supplementary Figs. 7 and 9). This interelectrode thickness (~0.3 mm, over five times the thickness of the AEM) was necessary to prevent a short circuit.

The membrane-free system did have low hydrogen-evolution reaction (FE < 5%) and good selectivity towards C_{2+} products (FE_{C2+} > 90%, Supplementary Fig. 10). AEMs, while not perfectly selective, serve to block alkali cation migration to the cathode catalyst. In contrast, in the membrane-free system, bulk electrolyte readily permits the passage of solvated cations, and these can build up at the interface between the hydrophobic support and the hydrophilic ionomer-coated catalyst layer, forming a hydrophobic:hydrophilic abrupt junction. We modelled the concentration of solvated cations at the junction, finding it to reach 4.5 M for the membrane-free approach, double that of the AEM case (Fig. 1d). The establishment of a cation superconcentration near the cathode may contribute to high C_{2+} productivity[15,26,27].

Development and optimization of porous separator for COR

We sought to preserve this benefit, but to block gas crossover, something we would seek to do with the use of a porous hydrophilic membrane inspired by alkaline water electrolysers (AWEs)[28] and recent

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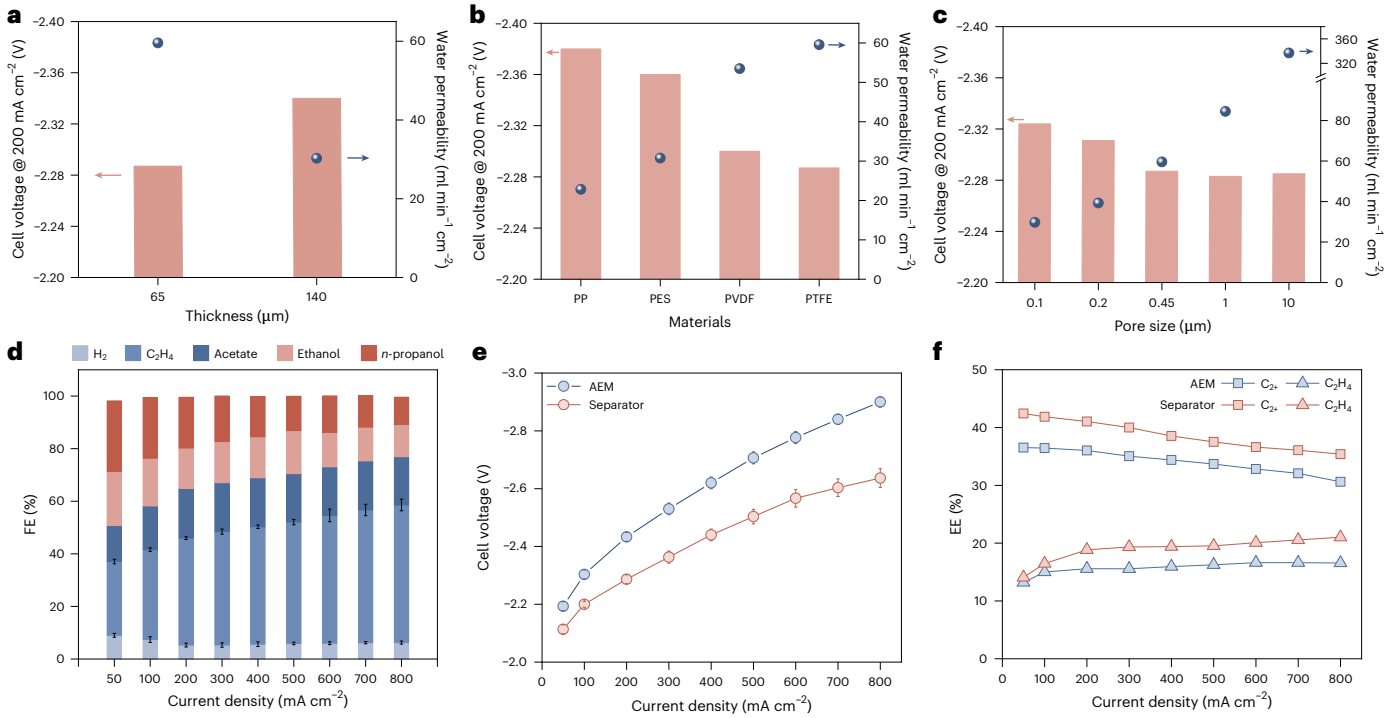

Fig. 2 | Performance of the MEA with a separator. a–**c**, Cell voltage at 200 mA cm^{-2} for different separator thicknesses (**a**), different separator materials (**b**) and different separator pore sizes (**c**). PP, polypropylene; PES, polyethersulfone; PVDF, polyvinylidene fluoride; PTFE, polytetrafluoroethylene. **d**, FE of the MEA with a separator. **e**,**f**, Cell voltage (**e**) and EE (**f**) of the MEA with an AEM and a separator at different current densities. Where error bars are shown, values are means, and error bars indicate s.d. (n = 3 replicates).

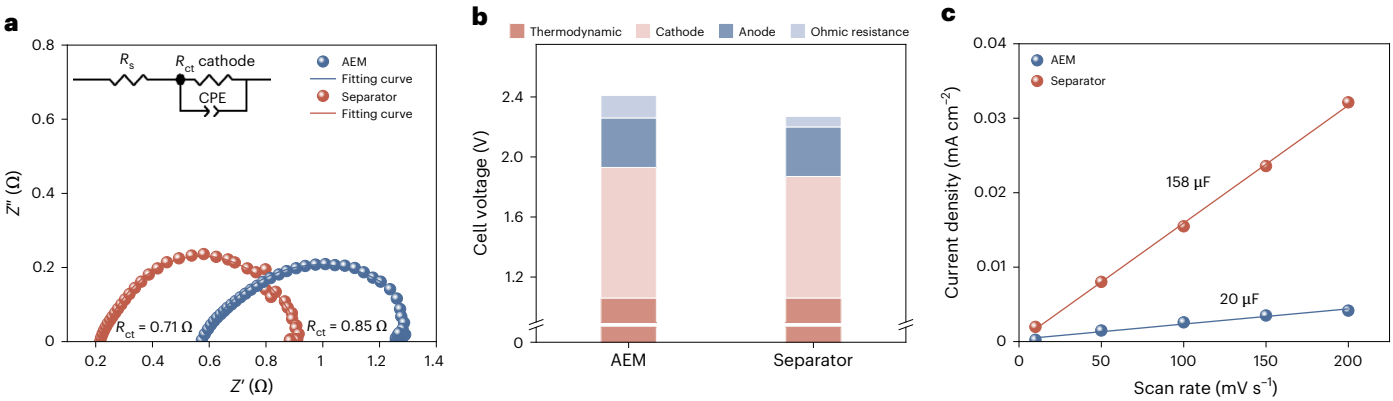

Fig. 3 | Mechanistic understanding of separator performance. a, Nyquist plot from EIS of the cathode versus the reference electrode in the MEA with an AEM and a separator, recorded at 200 mA cm$^{minus;2}$. CPE, constant phase element. **b**, Voltage breakdown of the MEA with an AEM and a separator, recorded at 200 mA cm$^{minus;2}$. **c**, Linear relation between the double-layer charging current and the scan rate, used to determine the electrochemical surface areas for the MEA with an AEM and a separator.

applications in CO_2R[22,29,30]. This porous separator simultaneously provides three functions: physically separating electrodes, maintaining effective gas-blocking capability by establishing a stable water-filled hydrophilic barrier to gas transport, and facilitating rapid ion transport through its highly porous and thin structure, thereby reducing ionic resistance and cell voltage (Fig. 1e).

When we tested a range of standard separators including ZIRFON—the benchmark for AWE—with thicknesses of ~200–300 μm and porosities of ~40–60% in COR (Supplementary Table. 3 and Supplementary Fig. 11), unfortunately we found that no appreciable voltage benefit was accrued relative to AEM COR, nor was the C_{2+} FE higher. These findings agree with previous CO_2R reports using ZIRFON, which reported only modest[30] or no improvement[29] in voltage.

We noted the chief reason why AWE employs separators in this regime of thickness and porosity: in the hydrogen-evolution reaction, dissolved H_2 diffuses rapidly through water-filled pores, and the field focuses thus on reducing water permeability (achieved through the use of small pores) and on increasing thickness to slow H_2 diffusion.

In contrast, in COR to C_2H_4, both feedstock and product diffuse much more slowly in water. Even when using a thin and highly porous separator, we found that gas product crossover was similar to that in AEM COR (Fig. 1f), with each approach providing C_2H_4 below 0.1 vol.% on the anode. We investigated how separator thickness, material composition and pore size affect COR performance (Fig. 2a–c and Supplementary Figs. 12–14). These results reveal clear trends: cell voltage and C_{2+} FE improved with thinner and more porous materials.

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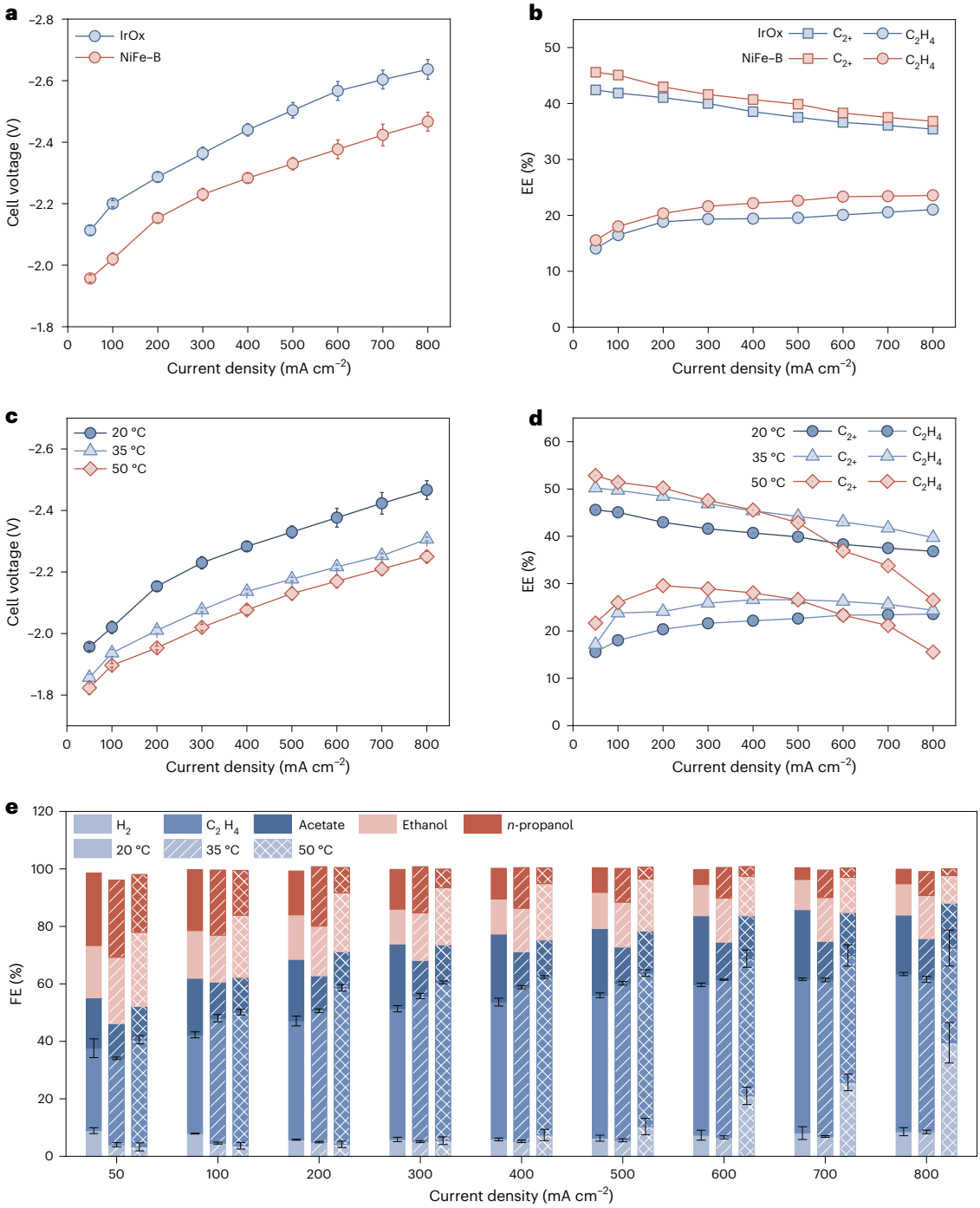

Fig. 4 | Optimization of the MEA with a separator. a,**b**, Cell voltage (**a**) and EE (**b**) of the MEA with a separator using IrOx and NiFe–B as the anode. **c**,**d**, Cell voltage (**c**) and EE (**d**) of the MEA with a separator at different operating temperatures. **e**, Product distributions of the MEA with a separator at different operating temperatures. Where error bars are shown, values are means, and error bars indicate s.d. ($n = 3$ replicates).

To establish a key descriptor for the separator in CO-to-C_{2+} electrolysis, we correlated performance trends with measured water permeability (Fig. 2a–c). Higher permeability shortened hydrated ion transport pathways and enabled greater cation concentration at the catalyst, contributing to improved selectivity and reduced cell voltage. Specifically, a separator with ~50–100-µm thickness, pore sizes of >0.45 µm and porosity of >80% (see scanning electron microscopy images in Supplementary Fig. 15) resulted in the lowest full-cell voltages and highest C_{2+} partial current densities (Fig. 2a–c and Supplementary Figs. 12 and 14). However, performance plateaued beyond a threshold water permeability, indicating that ion migration toward the cathode was no longer the limiting factor.

With the thin and highly porous separator and the abrupt hydrophobic:hydrophilic junction, the system achieved a C_{2+} FE of 95% and H_2 FE below 5% at current densities beyond 200 mA cm^{-2} (Fig. 2d and Supplementary Fig. 16). The production rate of liquid oxygenates depended on the supporting electrolyte cation, with $Li^+ < Na^+ < K^+$ (Supplementary Fig. 17) and Na^+ achieving the highest C_2H_4 FE. We therefore focused on Na^+ and varied NaOH concentration. We found that, although higher NaOH concentrations reduce

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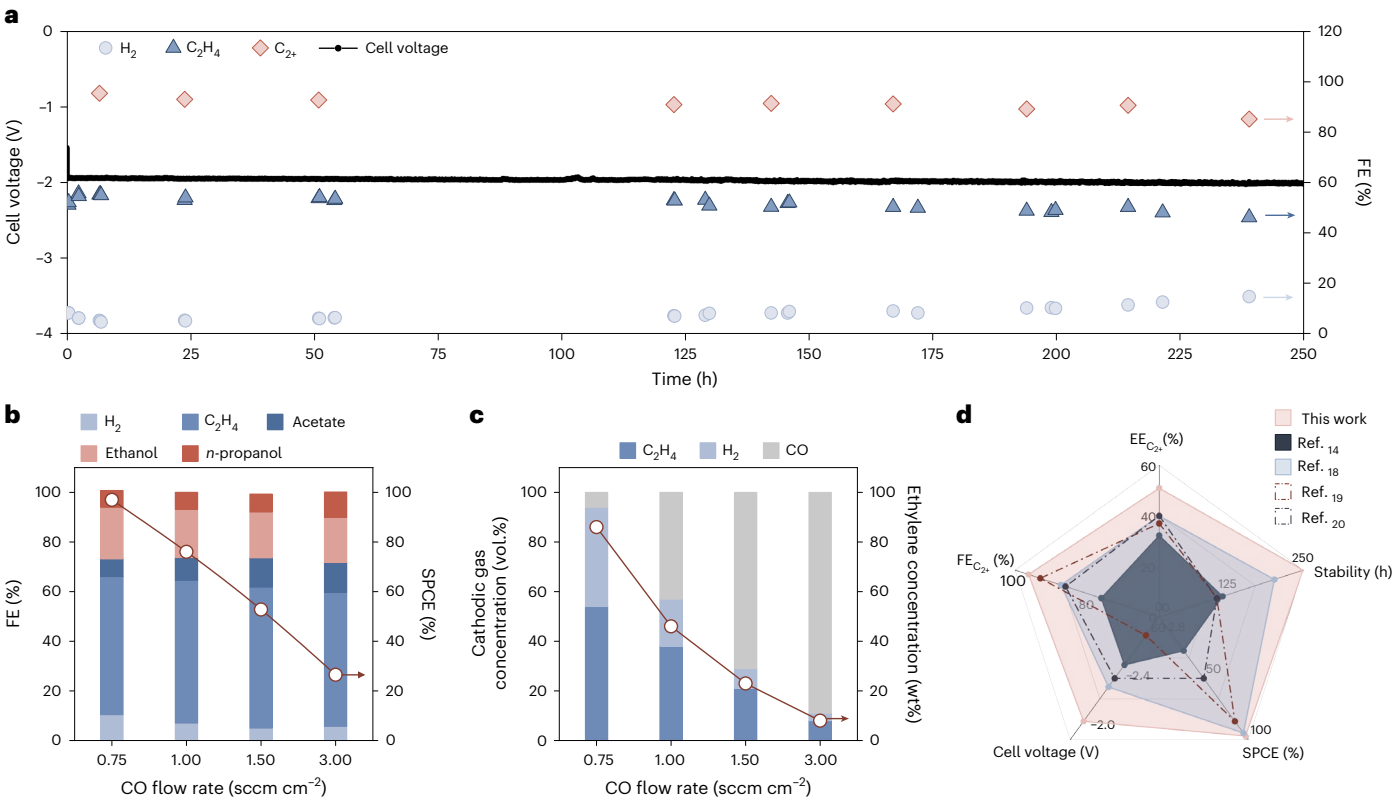

Fig. 5 | Stability and single-pass conversion of the MEA with a separator. a–c, Stability (**a**), single-pass conversion (**b**) and cathode gas product concentration (**c**) of the MEA with a separator. **d**, Comparison of EE, FE, cell voltage, stability and SPCE for C_{2+} production of the system in this work with those of state-of-the-art COR systems[14,18–20].

full-cell voltage due to improved conductivity, this benefit was offset by a decrease in C_{2+} FE and an increase in H_2 production beyond 2 M NaOH (Supplementary Fig. 18). We therefore used 2 M NaOH in ensuing studies.

In contrast with the large gap in the membrane-free cell, and also with the 200–300-μm gap of AWE separators, the separator enabled an ~60-μm separation between the electrodes. At 200 mA cm^{-2}, this led to a full-cell voltage of 2.28 V, a 150-mV reduction compared with the AEM case (Fig. 2e). This resulted in a C_2H_4 EE of 19% and a C_{2+} EE of 41% (Fig. 2f). The voltage reduction benefit increased with current density, offering a >300-mV improvement at 1 A cm^{-2}.

We investigated the origin of the reduced voltage by performing full-cell electrochemical impedance spectroscopy (EIS, Supplementary Fig. 19) to measure series resistance, R_s. We observed that R_s decreased from 0.8 to 0.4 Ω cm^2 when we switched to the separator. However, this reduction alone did not fully account for the overall voltage reduction achieved with the system—a decrease of 0.4 Ω cm^2 would correspond to a voltage saving of 80 mV at 200 mA cm^{-2}. We also saw a reduction in charge transfer resistance, R_{ct}. To isolate the voltage contributions from each electrode component, we incorporated a reference electrode and performed half-cell EIS measurements (cathode versus reference)[31]. Our analysis showed that the reduction in R_{ct} occurred at the cathode (Fig. 3a). We further resolved all internal voltages and found that the separator reduced ohmic resistance by approximately 80 mV and lowered the cathode overpotential by about 60 mV (Fig. 3b). We attribute the latter improvement to cation super-concentration at the cathode, which enhances activity and reduces the kinetic barrier to C_{2+} product formation.

We further measured the electrochemical active surface area, looking for indications that a catalyst directly wetted by electrolyte through the hydrophilic domains could enhance the triple phase boundary. We measured double-layer capacitances via cyclic voltammetry scans in the non-Faradaic potential range (Fig. 3c and Supplementary Fig. 20) and found the electrochemical active surface area to be 8 cm^2 compared with 1 cm^2 for the AEM.

Enhancing performance through anode catalyst and temperature optimization

We turned to the anode catalyst to further lower the full-cell voltage. Earth-abundant Ni-based catalysts have been reported to outperform Ir-based catalysts for alkaline OER[32–34]. We therefore replaced IrOx with a nickel–iron–boride anode (NiFe–B) (Supplementary Figs. 21–25). The product distribution from COR was unaltered (Supplementary Fig. 26). Operating at 200 mA cm^{-2}, the NiFe–B catalyst enabled a full-cell voltage of 2.15 V (Fig. 4a) and an increased C_2H_4 EE of 20% and C_{2+} EE of 43% (Fig. 4b). A voltage distribution analysis showed that NiFe–B reduced the anodic overpotential by 100 mV (Supplementary Fig. 27), and the half-cell EIS further showed that the catalyst enhances charge transfer, providing accelerated OER kinetics under alkaline conditions (Supplementary Fig. 27).

The thermal stability of the separator allows for experiments at elevated temperatures—conditions that would accelerate chemical degradation in conventional AEMs[35,36]. We varied operating temperature by adding an external heat source (Fig. 4c–e). Elevated operating temperatures can increase EE by accelerating kinetics of both the cathodic COR and anodic OER and increasing the conductivity of the electrolyte[37]. At higher temperatures, we observed first a reduction in H_2 FE coinciding with a slight improvement in C_2H_4 FE and C_{2+} FE to over 55% and 96%, respectively (Fig. 4e and Supplementary Fig. 28). The peak C_2H_4 FE and C_{2+} FE shifted towards lower current densities with increased temperature, a finding we attribute to decreased CO solubility and more optimal CO coverage for C_{2+} generation[38–40]. Ultimately, we achieved a

full-cell voltage (Fig. 4c) of 1.95 V and a C_{2+} EE of 51% at 200 mA cm^{-2} by optimizing to 50 °C (Fig. 4d and Supplementary Table 5). We found that the elevated temperature primarily reduced the cathodic overpotential while also slightly lowering the ohmic resistance and anodic overpotential (Supplementary Fig. 29). Through optimization, we achieved a cumulative voltage reduction of ~500 mV at 200 mA cm^{-2} compared with the baseline AEM COR case (Supplementary Fig. 30).

Long-term stability and CO single-pass conversion

Achieving extended operating stability has been a challenge in COR systems. We conducted prolonged electrolysis galvanostatically at 200 mA cm^{-2} and 50 °C (Fig. 5a). The system remained stable for 250 operating hours, with EE decaying by less than 10% by the end of the stability run (Supplementary Table 4). To further challenge the stability of the separator material, we reused the separator after the stability run with new electrodes. The performance was comparable to that with a fresh one, indicating that even after long run times the separator is not limiting (Supplementary Fig. 31).

We then sought to increase CO single-pass conversion efficiency (SPCE) by tuning the operating conditions of the electrolyser (Fig. 5b). High CO SPCE increases product concentration at the outlet, decreasing the downstream separation energy[41,42]. At 200 mA cm^{-2} at a flow rate of 0.75 sccm cm^{-2}, we achieved SPCE of 97% towards C_{2+} products (Fig. 5b), with the C_2H_4 FE and C_{2+} FE maintained at 56% and 90%. The result is a C_2H_4 concentration at the outlet of 87 wt% (or 54 vol.%, Fig. 5c). The porous separator and resulting hydrophobic:hydrophilic abrupt junction contribute to these improvements in FE, EE, SPCE and stability towards C_{2+} products (Fig. 5d and Supplementary Table 5).

Conclusion

We find that a porous hydrophilic domain blocks gas crossover, allows rapid ion conduction and provides a cation superconcentration that accelerates COR. This approach increases EE, prolongs operating stability and supports high SPCE. This work highlights the potential for microporous components to improve upon charge-selective membranes in electrolysers. Beyond the COR electrolyser, other systems limited by ion transport could benefit, particularly those with anode and cathode chambers that differ in phase (for example, a gas-fed cathode and liquid-fed anode as here, or the reverse). The robustness of the separator material offers further opportunities to operate in harsh conditions, such as high temperatures, strongly acidic or alkaline electrolytes, or the presence of produced alcohols that can degrade conventional membrane materials. The performance, simplicity and durability of this approach advance CO_2 and CO conversion toward industrial application.

Methods

Synthesis of the NiFe–B catalyst

The NiFe–B catalyst was synthesized using a method reported previously[34]. Briefly, 332.8 mg of $NiCl_2{\cdot}6H_2O$ (99.9%, Sigma-Aldrich) and 227.1 mg of anhydrous $FeCl_3$ (≥99.99%, Sigma-Aldrich) were dissolved in 2 ml of deionized water. In a separate vial, 2 ml of 5 M $NaBH_4$ (≥98%, Sigma-Aldrich) was prepared in deionized water. All solutions were cooled in an ice bath for 10 min to prevent uncontrolled hydrolysis to avoid formation of precipitates. The Ni and Fe precursor solution was then added dropwise to the $NaBH_4$ solution. The resulting black powder was then washed and centrifuged with water and ethanol three times each before vacuum drying.

Electrode preparation

Cathodes were prepared by spray-coating a Cu ink onto hydrophobic carbon paper. To prepare the ink, 75 mg of Cu nanoparticles (25-nm particle size, Sigma-Aldrich) and 250 µl of Nafion solution (5 wt% in mixture of lower aliphatic alcohols and water, contains 45% water, Sigma-Aldrich) were dispersed in 5 ml of methanol. The ink mixture was then sonicated for at least 1 h before being spray-coated on a 5 × 5 cm^2 hydrophobic carbon paper (Sigracet 39 BB, Fuel Cell Store). The cathode was then air-dried overnight before use. Before the test, the cathode was cut to 1 × 1 cm^2 for measurement.

For the stability test, homemade hydrophobic carbon paper was used instead. The homemade hydrophobic carbon paper was made by blade-casting a 0.5-mm layer of mixed PTFE and carbon nanoparticle ink over a 15 × 20 cm^2 carbon paper gas diffusion layer (AvCarb MGL 190 with 50% wet proof, Fuel Cell Earth). The carbon ink was prepared by mixing 0.4 g of carbon nanoparticles with 35 ml of isopropanol and four drops of Triton X-100 (Sigma-Aldrich). The PTFE dispersion was prepared by mixing 4 g of PTFE preparation (60 wt% dispersion in H_2O, Sigma-Aldrich) and 6 g of deionized water. Before the blade-casting, the carbon ink and the PTFE dispersion were each vortexed for 1 min, then mixed and stirred for an additional 40 s. After blade-casting, the resulting substrate was air-dried for 3 h before being placed in an oven for 3 h at 280 °C and 1 h at 340 °C. The resulting substrate was then used as the carbon paper for the stability test.

IrOx anodes were purchased from Magneto Special Anodes (Ir-mixed metal oxide (MMO)-coated titanium felt), cut to 1 × 1 cm^2 and used directly. To prepare the NiFe–B anode, 40 mg of the synthesized NiFe–B powder and 133 µl of Nafion solution (5 wt%) were dispersed in 3 ml of methanol. The ink mixture was then sonicated for at least 1 h. The resulting dispersion was then spray-coated onto a 2.5 × 5 cm^2 IrOx anode.

Electrolyser configuration

All experiments were performed in a customized electrolyser (316L stainless-steel cathode plate and grade 2 titanium anode plate) with an active area of 1 cm^2 accessed with a serpentine channel. To prevent liquid leakage around the electrode edges into the flow channel, the cathode electrode was securely affixed to the flow plate using adhesive tape. The MEA experiments were performed with an AEM (Sustainion X37-50 Grade RT, Dioxide Materials), which was sandwiched between the cathode and the anode. Both the cathode and the anode were protected and surrounded by a 0.01-in-thick silicone gasket to provide leak proofing. For the systems with separators, the set-up remained the same except that the AEM was replaced by a hydrophilic separator placed between the cathode and the anode. Unless otherwise specified, 20 sccm of CO (high purity, 2.5 grade, Praxair or Linde) was fed into the cathode while the anode was circulated with an electrolyte at a rate of 5 ml min^{-1}. Heating pads were used to heat the electrolyser for experiments with elevated temperatures. The temperature was measured using a thermocouple and was controlled by a proportional–integral–derivative controller.

Separators PTFE-0.1 (Omnipore, hydrophilic, 0.1-µm pore size), PTFE-0.45-thin (Omnipore, hydrophilic, 0.45-µm pore size), PTFE-0.45-thick (Millipore, hydrophilic, 0.45-µm pore size), PTFE-10 (Omnipore, hydrophilic, 10-µm pore size), PVDF-0.45 (Durapore, hydrophilic, 0.45-µm pore size), PES-0.45 (Millipore Express, hydrophilic, 0.45-µm pore size) and PP-0.45 (hydrophilic, 0.45-µm pore size) were purchased from Sigma-Aldrich. PP-Celgard (3419S microporous and 3420 non-woven membrane) was purchased from Celgard. Zirfon-UTP220 was purchased from Agfa. The detailed properties of all these separator materials are listed in Supplementary Table 3.

Electrochemical measurement

All electrochemical measurements were carried out using an electrochemical workstation (Autolab PGSTA204) connected to a current booster (Metrohm Autolab, 10 A). All voltage was reported directly without current–resistance compensation unless otherwise stated. Gas products were collected directly from the electrolyser cathode outlet using a gas-tight syringe and analysed with a gas chromatograph (PerkinElmer Clarus 590) coupled with a thermal conductivity detector and flame ionization detector. The liquid products were collected

separately from the cathode outlet and anolyte, respectively, and quantified with proton nuclear magnetic resonance spectroscopy using a water suppression model, with dimethylsulfoxide as the internal standard.

The following equations were used to determine the FE for the gas and liquid products, respectively:

$$\mathrm{FE}_{\mathrm{gas}} = x_i v_{\mathrm{g}} \frac{z_{\mathrm{p},i} FP}{RT} \frac{1}{I_{\mathrm{total}}} \times 100\%$$

$$\mathrm{FE}_{\mathrm{liquid}} = n_i \frac{z_{\mathrm{p},i} F}{Q} \times 100\%$$

where x_i represents the volume fraction of gas product i, v_{g} represents the gas flow rate at the outlet in sccm, $z_{\mathrm{p},i}$ represents the number of electrons required to produce one molecule of product i, F represents the Faraday constant, P represents the atmospheric pressure in Pa, R represents the ideal gas constant, T represents the temperature and I_{total} represents the total current; n_i represents the number of moles of product i and Q represents the charged passed while the liquid products are being collected.

EE was calculated as follows:

$$\mathrm{EE} = \sum_i^n \frac{E_{\mathrm{thermo},i}}{E_{\mathrm{cell}}} \mathrm{FE}_i$$

where $E_{\mathrm{thermo},i}$ represents the thermodynamic potential of product i and E_{cell} represents the full-cell voltage (non-current–resistance-compensated) of the electrolyser.

SPCE was calculated as follows:

$$\mathrm{SPCE} = \sum_i^n \frac{I_{\mathrm{total}} \frac{\mathrm{FE}_i}{z_{\mathrm{r},i} F} \times 24.5 \times 10^3 \left(\frac{\mathrm{cm}^3}{\mathrm{mol}}\right) \times 60 \left(\frac{\mathrm{s}}{\mathrm{min}}\right)}{v_{\mathrm{CO,input}}}$$

where $z_{\mathrm{r},i}$ represents the number of electrons required to react one molecule of CO for product i and $v_{\mathrm{CO,input}}$ represents the CO input flow rate in sccm.

Overpotentials of each electrode component were measured using an Ag/AgCl (3.5 M KCl) reference electrode positioned inline at the anolyte inlet of the electrolyser. The measured potentials were converted to the reversible hydrogen electrode scale and current–resistance-corrected using the following equation:

$$V(\mathrm{vs\ RHE}) = V(\mathrm{vs\ Ag/AgCl}) + 0.205 + 0.059\,\mathrm{pH} - iR.$$

Data availability

All data supporting the findings of this study are available within the paper and Supplementary Information files. Source data are provided with this paper.

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Author contributions

E.H.S. and D.S. supervised the work. R.K.M. conceived the idea. R.K.M. designed and carried out all the experiments with help from M.F., N.W. and Y.Z. R.K.M. prepared the manuscript. N.W. synthesized the NiFe–B anode material. F.L. performed the one-dimensional simulation. F.A. designed the electrolyser incorporating a reference electrode. C.P.O., P.P. and A.S.Z. prepared the hydrophobic carbon substrate. Y.L., W.N., K.X., Y.C., P.S., J.E.H. and J.W. assisted with experiments. M.L., J.K., Y.C.X., Y.X. and C.-T.D. contributed to data analysis and discussions. All authors discussed the results and approved the final version.

Competing interests

There is a US provisional patent application (US 63/656,427) titled "Interposed electrode assembly for electroreduction of carbon oxides to C_{2+} products", filed by R.K.M., D.S. and E.H.S. and their institutions. The remaining authors declare no competing interests.

Additional information

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